

Nifty 50 Behaviour Analysis

Do stocks move in a repeatable way around their corporate events — and can we turn that into a disciplined “buy-before / sell-after” framework?

A data-driven study of the 50 Nifty stocks • a live web dashboard • an honest set of findings

Data: NSE India | Presented by _____ • ____ / ____

THE IDEA

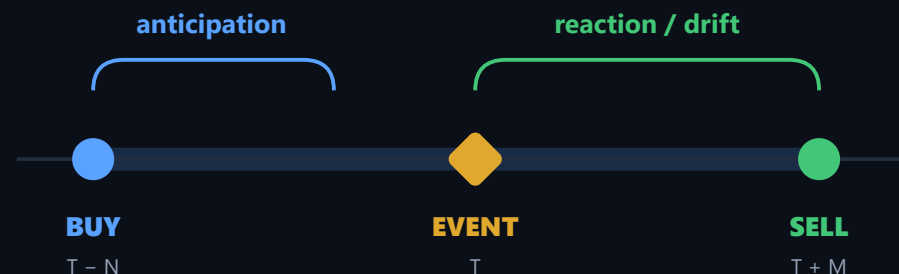
Trade the calendar, not a hunch.

Corporate events — **quarterly results, board meetings, dividends, announcements** — have **dates that are public in advance**. Markets **anticipate** them on the way in and **digest** them on the way out.

So we test one disciplined rule: **buy N days before** a known event, hold through it, and **sell M days after** — and measure how often that actually made money.

We never forecast the event.

The date is read off the regulatory calendar. The only thing we study is the *price behaviour* around it.



THE FOUNDATION

A clean, verified dataset.

50

stocks — the exact current Nifty 50, verified 1:1 with NSE

68,589

unique corporate-event dates analysed

106,034

raw event records processed

50/50

with fundamentals & full price history



NSE India

The event calendar — results, board meetings, corporate actions, announcements.

Universe re-synced to the exact 50 • full history merged & audited • date-parsing, duplication and coverage bugs found and fixed.

HOW IT WORKS

Brute-force honesty.

For each event we test **every** combination of “buy N days before” × “sell M days after” (N,M = 1...8) — an **8 × 8 = 64-rule grid** — against the stock’s full history, and keep the best.

```
return% = ( close[T+M] / close[T-N] - 1 ) × 100
WIN if > 0    // per event

win rate = wins ÷ total events × 100
```

no machine learning Just exhaustive search + transparent statistics — every number traces to one formula.

The 64 candidate rules



Where the strongest historical patterns sit.

Top stocks by **board-meeting** win rate — how often the best buy/sell rule closed positive. Bars start at zero; the count (n) is shown for honesty.



Historical hit-rate (in-sample). Higher n = more reliable. See slide 08 for the honest caveat.

THE PRODUCT

A live, interactive dashboard.

Behaviour

Each stock's best buy/sell rule & win rate per event, with a full return grid.

Rankings

All 50 stocks ordered by win rate, per event type.

Upcoming

A watchlist of events still ahead — with each stock's historical edge.

Screener

Filter by ROCE, growth, debt, P/E to find quality & cheap names.

Compare

2–3 stocks side by side — fundamentals + behaviour.

Fundamentals

Interactive **3D charts** with hover tooltips; offline Excel export.

Runs two ways.

A live Python app on your machine, and an always-on static snapshot on GitHub Pages — the same 50 stocks, anywhere.

From raw feeds to a polished tool.

1 Locked the universe

Re-synced to the exact current Nifty 50 and verified it 1:1 against NSE.

2 Fixed & deepened the data

Merged full history (+2,860 event-dates), backfilled board meetings 4×, extended corporate actions from Yahoo, repaired date-parsing & duplication bugs.

3 Built the analysis engine

The 64-rule grid, win-rate rankings, screener & compare — cached for instant loads.

4 Made it usable & beautiful

Upcoming watchlist, interactive 3D charts with tooltips, polished UI, readable filings — and an ML experiment kept honestly separate.

THE HONEST BOTTOM LINE

What the numbers really say.

Past win-rates around events are real — but they are **in-sample** (a rule chosen on the same history it's scored on) and **modest** out of sample.

No robust, reliably tradeable edge survives out-of-sample.

Even a pooled, time-validated ML test on 6,000 events landed at ~coin-flip.

That is itself a **valuable, defensible result** — and it's presented transparently, not hidden.

The real deliverable

A **transparent framework** and a **clean, verified, fully-historical Nifty 50 dataset** — a solid base for deeper, more careful research.

■ green = grew vs last year ■ red = fell / loss

● LIVE NOW

See it for yourself.

The full dashboard — 50 stocks, all event analyses, interactive 3D charts — is running and shareable.

Local app

`localhost:8090`

Always-on (GitHub Pages)

`yashsharma1202.github.io/behaviour-analysis/nifty/`

Source

`github.com/Yashsharma1202`

Nifty 50 Behaviour Analysis · for study & education, not investment advice · Thank you.

● APPENDIX

All 50 stocks — by sector

Headline fundamentals + the best event play & win rate for every constituent, grouped by sector for easy navigation.

All 50 stocks — by sector

Banks 5

HDFCBANK

₹825 · P/E 16.7

HDFC Bank

ROCE — OPM — D/E — Pft **+8%**

Results	65%
Board mtg	69%
Corp action ★	71%
Announce	58%

ICICIBANK

₹1,403 · P/E 18.5

ICICI Bank

ROCE — OPM — D/E — Pft **+6%**

Results	62%
Board mtg	72%
Corp action ★	74%
Announce	56%

AXISBANK

₹1,328 · P/E 15.6

Axis Bank

ROCE — OPM — D/E — Pft **-6%**

Results	64%
Board mtg	61%
Corp action ★	74%
Announce	59%

KOTAKBANK

₹377 · P/E 19.4

Kotak Mahindra Bank

ROCE — OPM — D/E — Pft **-14%**

Results	57%
Board mtg	65%
Corp action ★	81%
Announce	59%

SBIN

₹1,037 · P/E 11.5

State Bank of India

ROCE — OPM — D/E — Pft **+7%**

Results	61%
Board mtg	65%
Corp action ★	71%
Announce	57%

Financial Services 6

BAJFINANCE

₹1,014 · P/E 33.2

Bajaj Finance

ROCE — OPM — D/E — Pft **+15%**

Results	61%
Board mtg	64%
Corp action ★	72%
Announce	60%

BAJAJFINSV

₹1,901 · P/E 31.0

Bajaj Finserv

ROCE **11%** OPM **38%** D/E **5.52** Pft **+12%**

Results	62%
Board mtg	59%
Corp action ★	76%
Announce	58%

SHRIRAMFIN

₹1,044 · P/E 19.6

Shriram Finance

ROCE — OPM — D/E — Pft **+22%**

Results	63%
Board mtg	62%
Corp action ★	67%
Announce	58%

JIOFIN

₹242 · P/E 98.5

Jio Financial Services

ROCE **2%** OPM **66%** D/E **0.16** Pft **-5%**

Results	50%
Board mtg	58%
Corp action ★	100%
Announce	53%

HDFCLIFE

₹570 · P/E 64.4

HDFC Life Insurance Company

ROCE **10%** OPM **1%** D/E **0.17** Pft **+15%**

Results	50%
Board mtg	66%
Corp action ★	90%
Announce	51%

SBILIFE

₹1,846 · P/E 75.0

SBI Life Insurance Company

ROCE **15%** OPM **1%** D/E **0.00** Pft **—**

Results	64%
Board mtg	68%
Corp action ★	100%
Announce	61%

IT & Technology 5

TCS

₹2,126 · P/E 15.6

Tata Consultancy Services

ROCE **63%** OPM **27%** D/E **0.11** Pft **+8%**

Results	61%
Board mtg ★	62%
Corp action	60%
Announce	59%

INFY

₹1,070 · P/E 14.7

Infosys

ROCE **40%** OPM **24%** D/E **0.10** Pft **+13%**

Results	58%
Board mtg	64%
Corp action ★	67%
Announce	61%

HCLTECH

₹1,174 · P/E 19.1

HCL Technologies

ROCE **31%** OPM **21%** D/E **0.07** Pft **+1%**

Results ★	68%
Board mtg	61%
Corp action	66%
Announce	61%

TECHM

₹1,465 · P/E 29.8

Tech Mahindra

ROCE **23%** OPM **16%** D/E **0.07** Pft **+27%**

Results	58%
Board mtg ★	70%
Corp action	68%
Announce	64%

WIPRO

₹176 · P/E 14.0

Wipro

ROCE **18%** OPM **19%** D/E **0.23** Pft **+1%**

Results ★	59%
Board mtg	57%
Corp action	52%
Announce	55%

Automobiles 5

MARUTI ₹13,632 · P/E 29.2 Maruti Suzuki India ROCE 19% OPM 12% D/E 0.00 Pft +2% Results61% Board mtg61% Corp action★70% Announce57% <td> M&M ₹3,093 · P/E 22.5 Mahindra & Mahindra ROCE 15% OPM 19% D/E 1.44 Pft +37% Results65% Board mtg★84% Corp action78% Announce55% <td> BAJAJ-AUTO ₹10,087 · P/E 26.2 Bajaj Auto ROCE 28% OPM 21% D/E 0.58 Pft +48% Results68% Board mtg67% Corp action★78% Announce63% </td></td>	M&M ₹3,093 · P/E 22.5 Mahindra & Mahindra ROCE 15% OPM 19% D/E 1.44 Pft +37% Results65% Board mtg★84% Corp action78% Announce55% <td> BAJAJ-AUTO ₹10,087 · P/E 26.2 Bajaj Auto ROCE 28% OPM 21% D/E 0.58 Pft +48% Results68% Board mtg67% Corp action★78% Announce63% </td>	BAJAJ-AUTO ₹10,087 · P/E 26.2 Bajaj Auto ROCE 28% OPM 21% D/E 0.58 Pft +48% Results68% Board mtg67% Corp action★78% Announce63%
EICHERMOT ₹7,296 · P/E 36.3 Eicher Motors ROCE 31% OPM 25% D/E 0.02 Pft +17% Results67% Board mtg★86% Corp action81% Announce58% <td> TMPV ₹327 · P/E 1.5 Tata Motors Passenger Vehicles ROCE 3% OPM 6% D/E 0.71 Pft -77% Results★86% Board mtg67% Corp action61% Announce54% <td></td></td>	TMPV ₹327 · P/E 1.5 Tata Motors Passenger Vehicles ROCE 3% OPM 6% D/E 0.71 Pft -77% Results★86% Board mtg67% Corp action61% Announce54% <td></td>	

Metals & Mining 4

TATASTEEL ₹188 · P/E 21.7 Tata Steel ROCE 13% OPM 15% D/E 0.90 Pft +217% Results57% Board mtg★63% Corp action55% Announce56% <td> JSWSTEEL ₹1,238 · P/E 13.6 JSW Steel ROCE 11% OPM 16% D/E 0.99 Pft +134% Results67% Board mtg71% Corp action★80% Announce59% <td> HINDALCO ₹963 · P/E 16.2 Hindalco Industries ROCE 14% OPM 13% D/E 0.73 Pft +9% Results53% Board mtg★92% Corp action75% Announce58% </td></td>	JSWSTEEL ₹1,238 · P/E 13.6 JSW Steel ROCE 11% OPM 16% D/E 0.99 Pft +134% Results67% Board mtg71% Corp action★80% Announce59% <td> HINDALCO ₹963 · P/E 16.2 Hindalco Industries ROCE 14% OPM 13% D/E 0.73 Pft +9% Results53% Board mtg★92% Corp action75% Announce58% </td>	HINDALCO ₹963 · P/E 16.2 Hindalco Industries ROCE 14% OPM 13% D/E 0.73 Pft +9% Results53% Board mtg★92% Corp action75% Announce58%
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COALINDIA

₹428 · P/E 8.5

Coal India

ROCE **35%** OPM **24%** D/E **0.12** Pft **-12%**

Results	58%
Board mtg ★	68%
Corp action	64%
Announce	62%

Oil, Gas & Power ⁴

RELIANCE

₹1,304 · P/E 21.8

Reliance Industries

ROCE **10%** OPM **17%** D/E **0.45** Pft **+14%**

Results	59%
Board mtg	62%
Corp action ★	72%
Announce	57%

ONGC

₹247 · P/E 7.5

Oil & Natural Gas Corporation

ROCE **14%** OPM **16%** D/E **0.47** Pft **+15%**

Results	50%
Board mtg ★	63%
Corp action	62%
Announce	56%

NTPC

₹347 · P/E 12.4

NTPC

ROCE **8%** OPM **28%** D/E **1.33** Pft **+20%**

Results	58%
Board mtg	60%
Corp action ★	76%
Announce	58%

POWERGRID

₹285 · P/E 16.6

Power Grid Corporation of India

ROCE **10%** OPM **75%** D/E **1.47** Pft **+4%**

Results	60%
Board mtg	56%
Corp action ★	78%
Announce	56%

Pharma & Healthcare 5

SUNPHARMA

₹1,920 · P/E 40.1

Sun Pharmaceutical Industries

ROCE 21% OPM 30% D/E 0.06 Pft +9%

Results

Board mtg

Corp action ★

Announce

68%

50%

73%

61%

CIPLA

₹1,431 · P/E 29.8

Cipla

ROCE 15% OPM 21% D/E 0.02 Pft -26%

Results

Board mtg

Corp action ★

Announce

56%

69%

77%

58%

DRREDDY

₹1,242 · P/E 24.7

Dr. Reddy's Laboratories

ROCE 14% OPM 19% D/E 0.20 Pft -24%

Results

Board mtg ★

Corp action

Announce

53%

80%

76%

58%

APOLLOHOSP

₹8,828 · P/E 65.4

Apollo Hospitals Enterprise

ROCE 18% OPM 15% D/E 0.90 Pft +40%

Results

Board mtg ★

Corp action

Announce

74%

80%

73%

66%

MAXHEALTH

₹1,084 · P/E 73.2

Max Healthcare Institute

ROCE 15% OPM 27% D/E 0.32 Pft +31%

Results

Board mtg

Corp action ★

Announce

88%

77%

100%

66%

FMCG 4

HINDUNILVR

₹2,136 · P/E 33.4

Hindustan Unilever

ROCE 28% OPM 23% D/E 0.03 Pft +46%

Results

Board mtg

Corp action ★

Announce

53%

62%

69%

56%

ITC

₹281 · P/E 17.0

ITC

ROCE 39% OPM 35% D/E 0.03 Pft +6%

Results

Board mtg

Corp action ★

Announce

57%

43%

69%

57%

NESTLEIND

₹1,442 · P/E 79.5

Nestle India

ROCE 85% OPM 23% D/E 0.09 Pft +11%

Results ★

Board mtg

Corp action

Announce

71%

62%

69%

71%

TATACONSUM

₹1,103 · P/E 70.8

Tata Consumer Products

ROCE **9%** OPM **14%** D/E **0.13** Pft **+22%**

Results	55%
Board mtg	53%
Corp action ★	74%
Announce	59%

Consumer & Retail 4

TITAN

₹4,565 · P/E 79.9

Titan Company

ROCE **26%** OPM **10%** D/E **0.93** Pft **+54%**

Results	61%
Board mtg	46%
Corp action ★	76%
Announce	63%

TRENT

₹2,898 · P/E 89.8

Trent

ROCE **28%** OPM **19%** D/E **0.37** Pft **+19%**

Results	57%
Board mtg	66%
Corp action ★	80%
Announce	66%

ASIANPAINT

₹2,648 · P/E 58.7

Asian Paints

ROCE **26%** OPM **19%** D/E **0.18** Pft **+13%**

Results	58%
Board mtg	59%
Corp action ★	67%
Announce	64%

ETERNAL

₹294 · P/E 774.1

Eternal

ROCE **3%** OPM **2%** D/E **0.15** Pft **-16%**

Results ★	77%
Board mtg	65%
Corp action	—
Announce	61%

Cement & Materials 2

ULTRACEMCO

₹11,594 · P/E 41.8

UltraTech Cement

ROCE 13% OPM 19% D/E 0.31 Pft +36%

Results

Board mtg

Corp action ★

Announce

54%

74%

80%

62%

GRASIM

₹3,193 · P/E 43.8

Grasim Industries

ROCE 8% OPM 21% D/E 2.20 Pft +38%

Results

Board mtg

Corp action ★

Announce

61%

61%

74%

58%

Infra & Capital Goods 2

LT

₹3,916 · P/E 33.5

Larsen & Toubro

ROCE 15% OPM 12% D/E 1.15 Pft +7%

Results

Board mtg ★

Corp action

Announce

61%

77%

66%

59%

BEL

₹410 · P/E 49.5

Bharat Electronics

ROCE 37% OPM 29% D/E 0.00 Pft +14%

Results

Board mtg ★

Corp action

Announce

56%

82%

67%

64%

Diversified & Ports 2

ADANIENT

₹3,156 · P/E 43.6

Adani Enterprises

ROCE 6% OPM 14% D/E 1.32 Pft -199%

Results

Board mtg

Corp action ★

Announce

65%

64%

67%

62%

ADANIPORTS

₹1,820 · P/E 32.7

Adani Ports and Special Economic Zone

ROCE 14% OPM 59% D/E 0.66 Pft +15%

Results

Board mtg ★

Corp action

Announce

66%

75%

57%

64%

Telecom 1

BHARTIARTL

₹1,905 · P/E 43.5

Bharti Airtel

ROCE 18% OPM 57% D/E 1.31 Pft +27%

Results	52%
Board mtg ★	73%
Corp action	65%
Announce	60%

Aviation 1

INDIGO

₹5,077 · P/E —

InterGlobe Aviation

ROCE 7% OPM 14% D/E 11.15 Pft -117%

Results	69%
Board mtg	75%
Corp action ★	78%
Announce	63%